

## Angles, Quadrilaterals & Polygons

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### Part A — Parallel Lines Cut by a Transversal

#### Problem 1

Given  $m\angle 1 = 65^\circ$ , find  $m\angle 2$ ,  $m\angle 5$ , and  $m\angle 8$ .

Vertical Angles

Corresponding Angles

Alternate Exterior Angles

$\angle 1$  and  $\angle 2$  are a **linear pair** (supplementary):

$$m\angle 2 = 180^\circ - 65^\circ = 115^\circ$$

$\angle 1$  and  $\angle 5$  are **corresponding angles** (equal when lines are parallel):

$$m\angle 5 = 65^\circ$$

$\angle 1$  and  $\angle 8$  are **alternate exterior angles** (equal when lines are parallel):

$$m\angle 8 = 65^\circ$$

$$m\angle 2 = 115^\circ, m\angle 5 = 65^\circ, m\angle 8 = 65^\circ$$

#### Problem 2

$\angle 4$  and  $\angle 6$ : type of angle pair, equal or supplementary, and  $m\angle 6$  if  $m\angle 4 = 112^\circ$ .

Consecutive Interior Angles

Co-Interior (Same-Side Interior)

(a)  $\angle 4$  and  $\angle 6$  are **consecutive (co-interior / same-side interior) angles** — they are on the *same* side of the transversal, between the parallel lines.

(b) They are **supplementary**:  $m\angle 4 + m\angle 6 = 180^\circ$

(c)  $m\angle 6 = 180^\circ - 112^\circ = 68^\circ$

**Consecutive interior (co-interior); Supplementary;  $m\angle 6 = 68^\circ$**

### Problem 3

Alternate interior angles:  $m\angle 3 = (2x + 10)^\circ$  and  $m\angle 5 = (3x - 20)^\circ$ . Find  $x$  and each angle.

Alternate Interior Angles

Solving Equations

Alternate interior angles are equal:

$$2x + 10 = 3x - 20$$

Subtract  $2x$  from both sides:  $10 = x - 20$

Add 20 to both sides:  $x = 30$

Check:  $m\angle 3 = 2(30) + 10 = 70^\circ$     $m\angle 5 = 3(30) - 20 = 70^\circ$  ✓

$$x = 30, m\angle 3 = m\angle 5 = 70^\circ$$

## Part B — Quadrilaterals

### Problem 4

Parallelogram  $ABCD$ ,  $m\angle A = 72^\circ$ . Find  $m\angle B$ ,  $m\angle C$ ,  $m\angle D$ .

Parallelogram

Opposite Angles  $\cong$

Consecutive Angles Supplementary

In a parallelogram, **opposite angles are equal**:  $m\angle C = m\angle A = 72^\circ$

**Consecutive angles are supplementary**:  $m\angle B = 180^\circ - 72^\circ = 108^\circ$

Opposite:  $m\angle D = m\angle B = 108^\circ$

Check:  $72 + 108 + 72 + 108 = 360^\circ \checkmark$

$m\angle B = 108^\circ$ ,  $m\angle C = 72^\circ$ ,  $m\angle D = 108^\circ$

### Problem 5

Trapezoid  $PQRS$ :  $m\angle P = 110^\circ$ . Find  $m\angle S$  on the same leg.

Trapezoid

Co-Interior Angles

In a trapezoid, angles on the **same leg** (between the parallel sides) are **co-interior (supplementary)**:

$$m\angle P + m\angle S = 180^\circ$$

$$m\angle S = 180^\circ - 110^\circ = 70^\circ$$

$m\angle S = 70^\circ$

### Problem 6

4 congruent sides + 4 right angles. Name it and circle all applicable names.

Square

Quadrilateral Hierarchy

(a) This is a **Square**.

(b) A square is *also* a rectangle (4 right angles  $\checkmark$ ), a rhombus (4  $\cong$  sides  $\checkmark$ ), a parallelogram (2 pairs parallel sides  $\checkmark$ ).

Trapezoid also applies under the *inclusive* definition (has  $\geq 1$  pair of parallel sides).

**(a) Square (b) Parallelogram, Rectangle, Rhombus, Square (and Trapezoid under inclusive def.)**

## Part C — Polygon Angle Sum Theorem

### Problem 7

Interior angle sum of a hexagon ( $n = 6$ ) using  $(n - 2) \times 180^\circ$ . Draw diagonals from one vertex.

Polygon Angle Sum

Triangle Method

$$\begin{aligned}(n - 2) \times 180^\circ &= (6 - 2) \times 180^\circ = 4 \times 180^\circ \\ &= 720^\circ\end{aligned}$$

Drawing diagonals from one vertex of the hexagon creates **4 triangles**. Each triangle has  $180^\circ$ , so  $4 \times 180^\circ = 720^\circ$ . ✓

**Interior angle sum =  $720^\circ$**

### Problem 8

Each interior angle of a regular octagon ( $n = 8$ ).

Regular Polygon

Each Interior Angle Formula

Total interior angle sum:  $(8 - 2) \times 180^\circ = 6 \times 180^\circ = 1080^\circ$

Divide by the number of angles (8):

$$\frac{1080^\circ}{8} = 135^\circ$$

**Each interior angle =  $135^\circ$**

### Problem 9

Interior angle sum =  $1260^\circ$ . Find  $n$  and name the polygon.

Reverse Angle Sum

Solving for n

$$(n - 2) \times 180 = 1260$$

$$\text{Divide both sides by 180: } n - 2 = \frac{1260}{180} = 7$$

$$\text{Add 2: } n = 9$$

A polygon with 9 sides is a **nonagon**.

**$n = 9$  — Nonagon**

### Problem 10

Each exterior angle of a regular pentagon ( $n = 5$ ). Bonus: each interior angle.

Exterior Angles

360° Rule

Interior + Exterior = 180°

All exterior angles of any polygon sum to  $360^\circ$ .

Each exterior angle (regular pentagon):  $\frac{360^\circ}{5} = 72^\circ$

**Bonus:** Each interior angle =  $180^\circ - 72^\circ = 108^\circ$

Confirm using formula:  $\frac{(5 - 2) \times 180^\circ}{5} = \frac{540^\circ}{5} = 108^\circ \checkmark$

**Each exterior angle =  $72^\circ$  ; Each interior angle =  $108^\circ$**